ENGINEERING HANDOFF / 30 OCTOBER 2026

Portfolio Website Development Report

A practical implementation handoff for another AI agent: current architecture, completed features, design and behavior decisions, validation, known gaps, and safe next steps.

Project: Pranav's personal portfolio
Workspace: C:\Users\Pranav Sai Sreekanta\OneDrive\Documents\Portfolio
Stack: Vite, React, TypeScript, Tailwind CSS, Framer Motion, Vitest

1. Project state at a glance

This is an existing single-page portfolio. Continue from its current implementation; do not scaffold a replacement or redesign the established Hero and scroll transition without an explicit request. The app currently renders the animated Hero, About section, and Projects section.

> **Current layout:** animated Hero and scroll transition -> About -> Projects. A persistent navigation is layered over the page and becomes visible after the Hero intro animation.

Build passed | Homepage responds | 7 / 7 tests passed | Not a Git repository

2. Technology and project configuration

- Vite + React + TypeScript with Tailwind CSS v4.
- Framer Motion provides scroll-linked transforms and animation.
- Vitest is used for the existing tests.
- The `@/` alias points to `src/` in both Vite and TypeScript configuration.
- Shared UI components live in `src/components/ui/`.
- Visual styling, responsive rules, and section layouts primarily live in `src/index.css`.

Commands: `npm run dev` starts Vite, `npm run build` runs TypeScript build checks and production build, and `npm test` runs Vitest.

3. Architecture and key files

- `src/main.tsx` mounts the React application and imports global CSS.
- `src/App.tsx` composes sections and coordinates the Hero-to-About scroll progress, character motion, background, and reduced-motion behavior.
- `src/components/ui/waving-portfolio-landing.tsx` contains the self-contained Hero art, letter storm, headline, waving character, dice/signature interactions, and profile links.
- `src/components/ui/container-scroll-animation.tsx` implements the scroll-driven transition and exposes its progress to App.
- `src/components/portfolio-nav.tsx` implements persistent nav, section links, active section tracking, intro visibility, corner dice, and smooth scrolling.
- `src/components/about-section.tsx` contains About heading, bio card, and character artwork.
- `src/components/projects-section.tsx` hosts featured projects and the technical-notes dialog.
- `src/components/ui/featured-project-hero.tsx` renders project selector tabs, preview, project detail, action links, and dialog trigger.
- `src/data/featured-projects.tsx` stores the project details separately from presentation.
- `src/components/project-previews.tsx` renders the editable project-image placeholder.

- `src/index.css` contains global, nav, About, Projects, and responsive styles; `src/App.test.tsx` contains the existing test suite.

4. Completed user-facing features

Hero and transition

- Hero configured for Pranav, 2026, roles Software Engineer and AI/ML.
- Intro sequence includes letter storm, PORTFOLIO headline motion, and character rise/wave.
- Interactive lettering, dice, and signature behaviors are part of the Hero component.
- Framer Motion scroll progress synchronizes the transition background, character movement/pose, and About content reveal. Reduced-motion preferences are considered.

Persistent navigation

- Fixed navigation surface/color is tied to transition progress, changing naturally from warm paper to black.
- Navigation visibility is delayed until the Hero intro completes. It tracks active sections and smooth-scrolls to existing anchors.
- Corner ornaments/dice mirror the Hero visuals. The mobile nav was compacted to avoid covering the About heading.

About

- Black background with Kanit typography: ABOUT filled off-white, ME outlined.
- Bio is inside a dark editorial card with restrained red detail and a small metadata header.
- Uses the shared character artwork, positioned alongside the card on larger viewports and scaled/positioned for mobile.
- Mobile layout is compressed to fit one viewport beneath the navbar.

Hero profile links

Resume, GitHub, and LinkedIn have red-outline circular styling inspired by the user-provided reference and stack vertically on narrow screens. Destinations are explicit placeholders, not invented links.

5. Projects implementation

Projects follows the portfolio's dark editorial visual language: near-black background, off-white text, restrained gray borders, red highlights, and compact responsive layout. The requested eyebrow/side copy were removed; the heading is smaller than the initial reference design.

1. **DataMind** - Agentic AI / Data Analytics. Python, LangGraph, LangChain, Gemini, FastAPI, React, Supabase / PostgreSQL. Describes stateful natural-language analysis, automated reporting, authentication, and persistence.
2. **Continuous Behavioral Biometric Authentication** - Machine Learning / Security. Python, feature engineering, Random Forest, scikit-learn. Reports ML Lead contribution, 30+ statistical features, and 91.21% accuracy, 97.37% ROC-AUC, 7.88% EER across 546 test pairs (as provided by the user).
3. **ResumeMatch** - AI Application / Full-Stack Development. React, FastAPI, semantic search, LLM APIs, rule-based matching. Highlights semantic matching and deterministic rules that reduce unnecessary LLM calls.

- Three accessible selectable tabs update the preview and details. Keyboard support includes arrow navigation, Home, and End.
- Each project has separate Live project and GitHub repo actions. Without configured URLs, the controls are disabled and identified as unconfigured.
- Read technical notes opens a native dialog populated with project-specific details.
- Project preview image placeholders specify target paths under `public/projects/`; no fabricated product screenshots are claimed.
- Responsive CSS stacks project details and preview, allows tab overflow handling, and wraps technology tags. Reduced-motion transitions are included.

6. Latest mobile layout fix and validation

The latest implementation change adjusted the mobile navbar and About spacing in `src/index.css`. The navbar now uses a compact three-column grid at widths up to 640px. About layout top padding accounts for the navbar; character scale and card spacing were reduced.

Browser checks reported the About heading below the navbar, the card contained within the viewport, and no character/card collision at these sizes:

- 320 x 640: nav about 99px high; heading y=112; card bottom y=630; no overlap.
- 375 x 667: nav about 99px high; heading y=112; card bottom y=634; no overlap.
- 390 x 844: nav about 107px high; heading y=112; card bottom y=681; no overlap.
- 1440 x 900: About heading and card fit; no character/card overlap.

Runtime validation plan completed successfully after dependency recovery:

- **Production build:** passed.
- **Homepage HTTP probe:** passed (HTTP 200).
- **Vitest:** 7 passed, 0 failed.

> **Environment caution:** An initial npm ci failed with Windows EPERM because Vite servers held a native dependency file open. Stopping identified project Vite processes allowed setup and validation to pass. Avoid starting duplicate dev servers; check for existing servers before npm ci.

7. Known gaps and items needing user input

- **Profile destinations:** actual Resume PDF, GitHub profile, and LinkedIn URLs are not configured.
- **Project destinations:** verified GitHub repositories and deployed live demos are not configured. Do not invent URLs or imply deployment.
- **Project visuals:** all project visuals remain placeholders. Actual images can replace them at `public/projects/datamind.png`, `public/projects/behavioral-biometrics.png`, and `public/projects/resumematch.png`.
- **Navigation destinations:** nav lists Skills, Certifications, Experience, and Contact, but App currently renders only About and Projects. Do not treat those extra anchors as implemented sections.
- **Year labels:** project entries display "Year not specified"; no project years were supplied.

8. Guidance for the next AI agent

- Implement the user's next requested change in the existing codebase; avoid unrelated files and visual redesigns.
- Preserve Hero animation/interactions, the scroll-driven Hero-to-About transition, persistent nav, and warm off-white/red/black palette.
- Do not fabricate links, assets, deployment status, project years, or results.
- When a behavior or scope choice materially affects design, ask for clarification instead of assuming.

- After code changes run `npm run build` and `npm test`; use responsive browser checks for layout changes.
- The workspace is not a Git repository; do not claim changes were committed.

Prepared as a handoff document from the current workspace and prior implementation notes. Reported test and responsive results reflect the latest recorded validation.